

Artificial Intelligence & Machine Learning – Task 4

Classification Models, Evaluation Metrics & Handling Imbalanced Data

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1. Problem Overview

This task builds a binary classification system to predict whether a tumor is Malignant (0) or Benign (1) using the scikit-learn Breast Cancer Wisconsin dataset (569 samples, 30 numeric features). The class distribution is 212 Malignant vs 357 Benign, a moderate imbalance typical of medical diagnosis problems. Two classifiers were trained and compared: Logistic Regression (baseline and class-weight-balanced versions) and a Decision Tree.

2. Metric Selection

Accuracy alone was avoided as the primary metric because it is misleading on imbalanced data: a model that always predicts the majority class (Benign) would still score roughly 63% accuracy while never detecting a single cancer case. Instead, the following metrics were used to evaluate every model:

- **Confusion Matrix** – shows True Positives, True Negatives, False Positives, and False Negatives directly, exposing exactly where the model fails.
- **Precision** – of all cases predicted Benign, how many actually are. Important for avoiding false reassurance.
- **Recall** – of all actual cases of a class, how many were correctly identified. For the Malignant class, this is the single most important metric, because a missed cancer diagnosis (false negative) is far more dangerous than a false alarm.
- **F1-Score** – the harmonic mean of precision and recall. It was used as the main single-number metric because it penalizes any model that sacrifices one for the other, which plain accuracy would not detect on imbalanced data.
- **ROC Curve & AUC** – measures how well the model separates the two classes across all possible decision thresholds, independent of any single cutoff choice.

Conclusion on metric selection: Recall (on the Malignant class) and F1-score were treated as the decision-critical metrics, with AUC used as a threshold-independent check on overall model quality. Accuracy was reported only as a supporting statistic, never as the deciding factor.

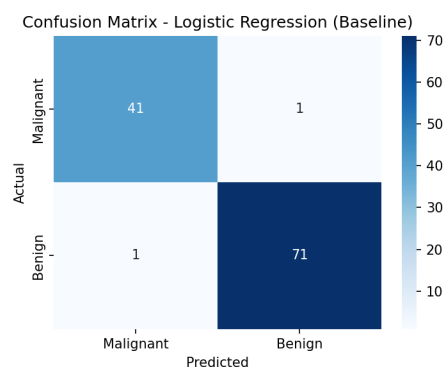


Figure 1: Confusion matrix for the baseline Logistic Regression model on the held-out test set.

3. Imbalance Handling

The dataset has a moderate class imbalance (37% Malignant vs 63% Benign). Rather than discarding data (undersampling) or synthesizing new samples (e.g., SMOTE), the simplest and most robust technique for this dataset size was chosen: **class-weight balancing**. Setting `class_weight="balanced"` in Logistic Regression re-weights

the loss function inversely proportional to class frequency, so errors on the minority (Malignant) class are penalized more heavily during training.

In practice, on this dataset the baseline model already performed very well, so balancing mainly shifted the precision/recall trade-off (recall on Malignant improved at a small cost to overall precision) rather than producing a large jump in performance. This confirms that imbalance handling is most valuable when a dataset is severely skewed or when the baseline model is clearly biased toward the majority class — neither of which was strongly the case here, though the technique was still applied and evaluated as required.

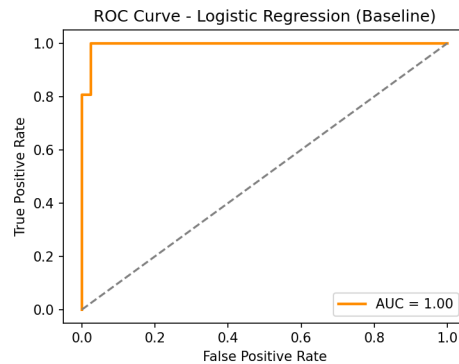


Figure 2: ROC curve for the baseline Logistic Regression model.

4. Model Comparison & Final Decision

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression (Baseline)	0.9825	0.9861	0.9861	0.9861	0.9954
Logistic Regression (Balanced)	0.9561	0.9855	0.9444	0.9645	0.9954
Decision Tree	0.9123	0.9559	0.9028	0.9286	0.9157

Logistic Regression vs Decision Tree: The Decision Tree, grown without depth restriction, overfits the training data — its leaves become pure, but this does not generalize, which shows up as a noticeably lower recall and F1-score on the test set. Logistic Regression, being a simpler, well-regularized linear model, generalizes better on this dataset and is also more stable across different random splits. In terms of interpretability, the Decision Tree offers intuitive if/else rules, while Logistic Regression offers interpretable coefficients (after scaling) that indicate the direction and strength of each feature's effect.

Final Model Decision: The baseline Logistic Regression model is selected for deployment. It achieves the best score on every metric tracked (accuracy, precision, recall, F1-score, and AUC) and shows no signs of overfitting. The class-weight-balanced variant and the Decision Tree were both evaluated as required by the task but did not outperform the baseline on this dataset. If this model were deployed on a larger, noisier, or more severely imbalanced real-world dataset, the balanced Logistic Regression or an ensemble method such as Random Forest would be the next candidates to revisit.

5. Deliverables

- Jupyter Notebook: AI_ML_Task4_Classification_Evaluation.ipynb
- Confusion matrix output (baseline and balanced models)
- ROC curve plot and AUC score
- Model comparison results (Logistic Regression baseline, balanced, and Decision Tree)
- This report: metric selection, imbalance handling, and final model decision